

ACTIVE CLUSTERING WITH NOISY PAIRWISE OBSERVATIONS

Karthik P N

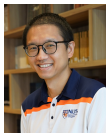
SPARC Workshop on Distributed Learning and Optimization, IISc

13 JULY 2026

Based on Joint Work With



Rachel Teo (NUS)

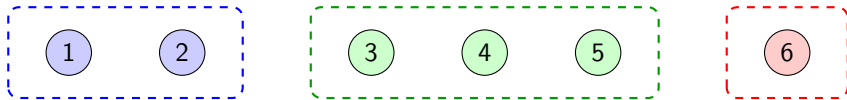


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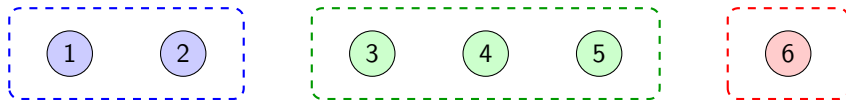
PROBLEM SETTING



- M items, each belonging to one of $K \in [M]$ clusters; K unknown

Example: $M = 6$; $K = 3$; $c_1 = c_2 = 1$, $c_3 = c_4 = c_5 = 2$, $c_6 = 3$

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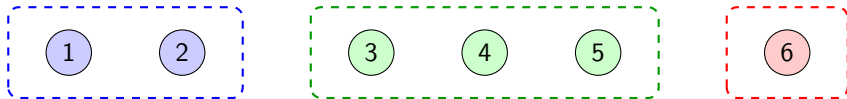
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- At each time $t = 1, 2, \dots$, query item pair $(i_t, j_t) = (i, j)$ to get

$$\mathbf{Y}_{ij} \sim v_{ij} = \begin{cases} \text{Ber}(p), & c_i = c_j \\ \text{Ber}(q), & c_i \neq c_j, \end{cases}$$

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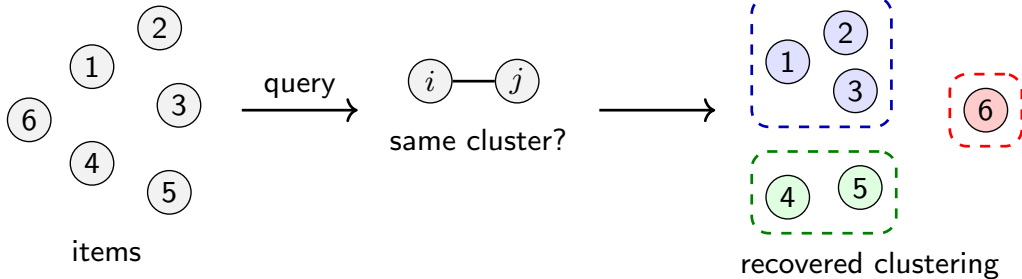
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- **Goal:** Given $\delta \in (0, 1)$, recover the underlying clustering correctly w.p. $\geq 1 - \delta$



INSTANCE MATRIX AND ADJACENCY MATRIX

INSTANCE C

	1	2	3	4	5	6
1	1	p	q	q	q	q
2	p	1	q	q	q	q
3	q	q	1	p	p	q
4	q	q	p	1	p	q
5	q	q	p	p	1	q
6	q	q	q	q	q	1

ADJACENCY MATRIX $C^=$

	1	2	3	4	5	6
1	1	1	0	0	0	0
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$C^{\neq}$

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- Our interest is only to recover $C^=$, not the instance matrix C
- The closer p and q are, the harder it is to recover $C^=$
- Two instances are **equivalent** if their adjacency matrices are identical

$$C' \sim C'' \iff C'^{=} = C''^{=}.$$

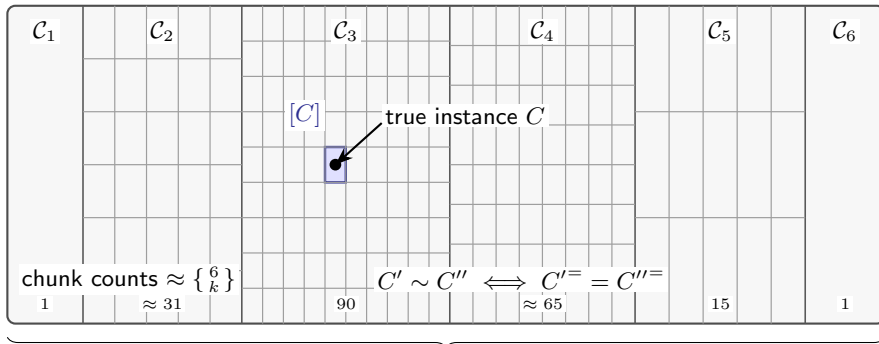
INSTANCE SPACE AND EQUIVALENCE CLASSES



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Instance Space $\mathcal{C}$

$\mathcal{C}_k$: instances with k clusters



M -fold partition by number of clusters $k = 1, \dots, M$

$$\mathcal{C} = \bigcup_{k=1}^M \mathcal{C}_k,$$

$$\# \text{chunks in } \mathcal{C}_k = \left\{ \begin{smallmatrix} M \\ k \end{smallmatrix} \right\}$$

Stirling number of the second kind

	Active Clustering (Our Setup)	Stochastic Block Model (SBM)
Observations	Actively query item pairs and receive noisy binary feedback; the same pair may be queried multiple times	Observe one random graph snapshot; each potential edge is sampled once through graph connectivity
Asymptotics	M fixed, $\delta \downarrow 0$ to zero	$M \rightarrow \infty$
Probabilities	p, q fixed and unknown	p, q scale like $\Theta\left(\frac{\log M}{M}\right)$

Takeaway: Ours is an adaptive, fixed-confidence sampling problem rather than passive recovery from a single random graph

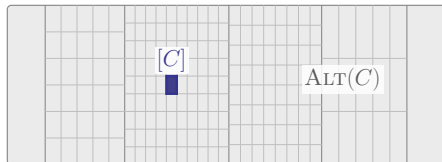
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- $\mathcal{P}_I$: probability distributions on I
- Underlying instance: C (unknown)
- The set of alternative instances is

$$\text{ALT}(C) = \bigcup_{\substack{C' \in \mathcal{C}: \\ [C'] \neq [C]}} [C'].$$



Theorem (Instance-Dependent Lower Bound)

Any δ -correct algorithm π satisfies

$$\mathbb{E}_C[\tau_\delta(\pi)] \geq \frac{\text{KL}(\text{Ber}(\delta) \parallel \text{Ber}(1 - \delta))}{D^\star(C)},$$

where

$$D^\star(C) = \sup_{\lambda \in \mathcal{P}_{\mathcal{I}}} \inf_{C' \in \text{ALT}(C)} \sum_{(i,j) \in \mathcal{I}} \lambda_{ij} \text{KL}(v_{ij}, v'_{ij}).$$

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- λ is the algorithm's sampling strategy
- **Learner:** choose λ to maximize the information gathered

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- For a fixed λ , **Nature** chooses the most confusing alternative C'
- The inner $\inf$ identifies the hardest alternative

Theorem (Instance-Dependent Lower Bound)

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- Weighted KL sum measures the distinguishability of C' from C under λ
- Learner vs. Nature: a max-min game whose value is $D^*(C)$

Theorem (Reduction to Minimal Alternatives)

We have

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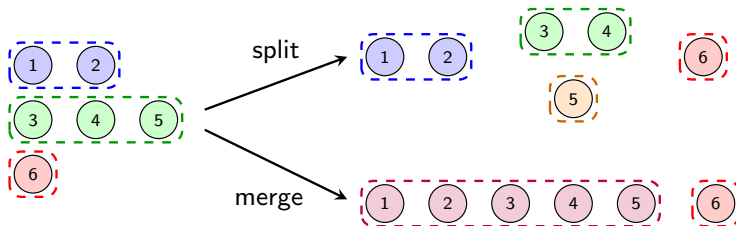
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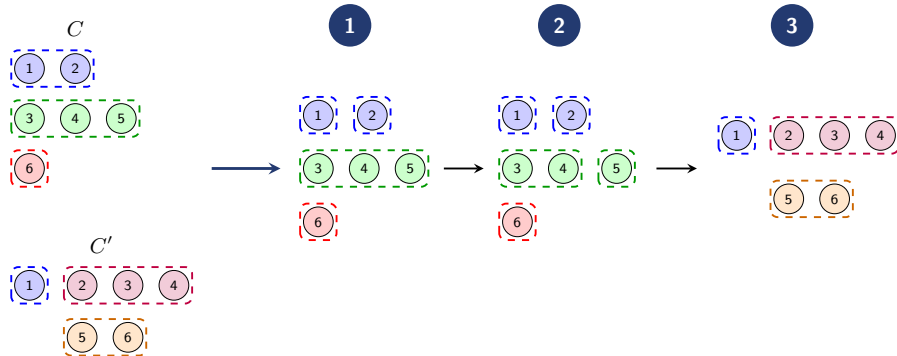
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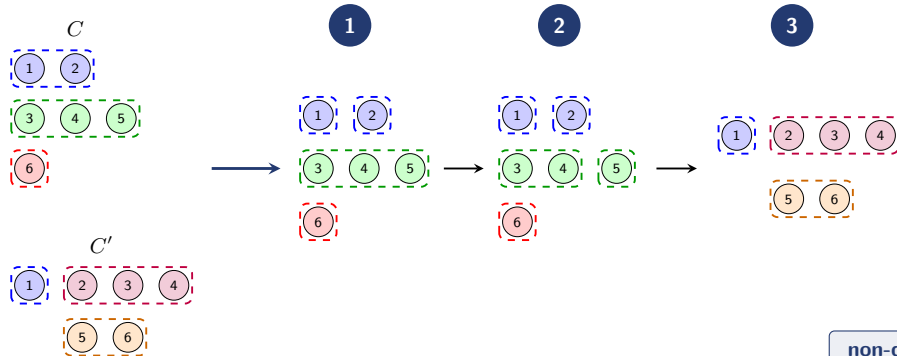


ALT(C) $\rightarrow$ MIN(C)



- $N_1(C, C') = \{(i, j) : c_i = c_j, c'_i \neq c'_j\} = \{(1, 2), (3, 5), (4, 5)\}$
 $N_2(C, C') = \{(i, j) : c_i \neq c_j, c'_i = c'_j\} = \{(2, 3), (2, 4), (5, 6)\}$

$$\text{ALT}(C) \rightarrow \text{MIN}(C)$$



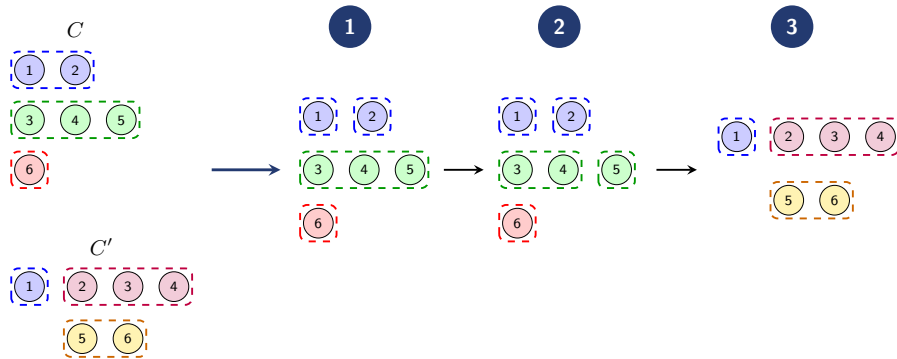
non-decreasing

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- The inner infimum can be written as

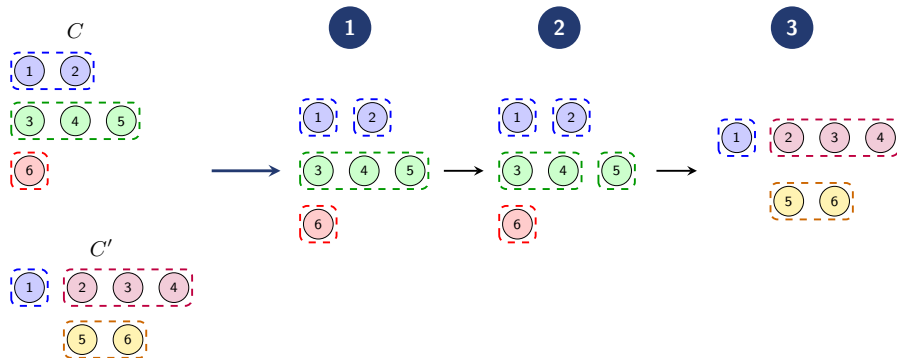
$$\inf_{C' \in \text{ALT}(C)} \sum_{(i,j) \in I} \lambda_{ij} \text{KL}(v_{ij}, v'_{ij}) = \min_{[C'] \in \text{ALT}(C)/\sim} z(\lambda_{N_1(C, C')}, \lambda_{N_2(C, C')}).$$

$$\text{ALT}(C) \rightarrow \text{MIN}(C)$$



- $N_1(C, C') = \{(1, 2), (3, 5), (4, 5)\},$ $N_1(C, \textcircled{1}) = \{(1, 2)\}$
 $N_2(C, C') = \{(2, 3), (2, 4), (5, 6)\},$ $N_2(C, \textcircled{1}) = \emptyset$
- $\lambda_{N_1(C, C')} = \lambda_{12} + \lambda_{35} + \lambda_{45},$ $\lambda_{N_1(C, \textcircled{1})} = \lambda_{12}$
 $\lambda_{N_2(C, C')} = \lambda_{23} + \lambda_{24} + \lambda_{56},$ $\lambda_{N_2(C, \textcircled{1})} = 0$

$$\text{ALT}(C) \rightarrow \text{MIN}(C)$$



$$\begin{aligned} N_1(C, C') &= \{(1, 2), (3, 5), (4, 5)\}, & N_1(C, \textcircled{1}) &= \{(1, 2)\} \\ N_2(C, C') &= \{(2, 3), (2, 4), (5, 6)\}, & N_2(C, \textcircled{1}) &= \emptyset \end{aligned}$$

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ALGORITHM: SAMPLING RULE



- Compute the empirical instance $\hat{C}_t$ given by

$$(\hat{C}_t)_{ij} = (\hat{C}_t)_{ji} = \frac{1}{N_{ij}(t)} \sum_{s=1}^t \mathbb{1}\{i_s = i, j_s = j\} \mathbf{Y}_{i_s j_s},$$

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- Compute the set of optimal allocations for $\hat{C}_t$:

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Why this does not work:

- C has only two levels (p, q) , but $\hat{C}_t$ need not

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- C has only two levels (p, q) , but $\hat{C}_t$ need not
- $\hat{C}_t \notin C$, and hence $\Lambda^*(\hat{C}_t)$ is not well-defined

EMPIRICAL PAIRWISE MEANS

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 (\hat{C}_t)_{ij} &= (\hat{C}_t)_{ji} \\
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 \end{aligned}$$

TYPICAL EMPIRICAL INSTANCE $\hat{C}_t$

	1	2	3	4	5	6
1	1	.78	.34	.22	.47	.18
2	.78	1	.41	.29	.36	.53
3	.34	.41	1	.72	.61	.44
4	.22	.29	.72	1	.67	.38
5	.47	.36	.61	.67	1	.27
6	.18	.53	.44	.38	.27	1

Many empirical levels, not just p, q

Solution: Project $\hat{C}_t$ onto $\mathcal{C}$ using $\text{Proj}_{\mathcal{C}}$, and set $C_t = \text{Proj}_{\mathcal{C}}(\hat{C}_t)$.

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6	.18	.53	.44	.38	.27	1

Many empirical levels, not just p, q

Solution: Project $\hat{C}_t$ onto $\mathcal{C}$ using $\text{Proj}_{\mathcal{C}}$, and set $C_t = \text{Proj}_{\mathcal{C}}(\hat{C}_t)$.

STEP 1: THRESHOLD AT $1/2$

$$(\hat{C}_t^=)_{ij} = \mathbb{1}\left\{(\hat{C}_t)_{ij} \geq \frac{1}{2}\right\},$$

$$(\hat{C}_t^{\neq})_{ij} = \mathbb{1}\left\{(\hat{C}_t)_{ij} < \frac{1}{2}\right\}.$$

for all $(i, j) \in I$.

EMPIRICAL MATRIX $\hat{C}_t$

	1	2	3	4	5	6
1	1	.78	.34	.22	.47	.18
2	.78	1	.41	.29	.36	.53
3	.34	.41	1	.72	.61	.44
4	.22	.29	.72	1	.67	.38
5	.47	.36	.61	.67	1	.27
6	.18	.53	.44	.38	.27	1

AFTER THRESHOLDING: $\hat{C}_t^=$

	1	2	3	4	5	6
1	1	1	0	0	0	0
2	1	1	0	0	0	1
3	0	0	1	1	1	0
4	0	0	1	1	1	0
5	0	0	1	1	1	0
6	0	1	0	0	0	1

$$\hat{C}_t^{\neq} = \mathbf{1} - \hat{C}_t^= \text{ entrywise}$$

Solution: Project $\hat{C}_t$ onto $\mathcal{C}$ using $\text{Proj}_{\mathcal{C}}$, and set $C_t = \text{Proj}_{\mathcal{C}}(\hat{C}_t)$.

EMPIRICAL MATRIX $\hat{C}_t$

	1	2	3	4	5	6
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6	.18	.53	.44	.38	.27	1

STEP 2: FIND THE NEAREST CLUSTERING TO $\hat{C}_t^=$

$$\tilde{C}_t = \arg \min_{M \in \mathcal{C} / \sim} \sum_{(i,j) \in I} \left| (\hat{C}_t^=)_{ij} - M_{ij}^= \right| + \left| (\hat{C}_t^{\neq})_{ij} - M_{ij}^{\neq} \right|$$

AFTER THRESHOLDING: $\hat{C}_t^=$

	1	2	3	4	5	6
1	1	1	0	0	0	0
2	1	1	0	0	0	1
3	0	0	1	1	1	0
4	0	0	1	1	1	0
5	0	0	1	1	1	0
6	0	1	0	0	0	1

$$\hat{C}_t^{\neq} = \mathbf{1} - \hat{C}_t^= \text{ entrywise}$$

Solution: Project $\hat{C}_t$ onto $\mathcal{C}$ using $\text{Proj}_{\mathcal{C}}$, and set $C_t = \text{Proj}_{\mathcal{C}}(\hat{C}_t)$.

STEP 3: ESTIMATE p_t AND q_t

ENTRIES AT LEAST $1/2$

$$p_t = \frac{\sum_{(i,j) \in I} \mathbb{1}\{(\hat{C}_t^=)_{ij} = 1\} (\hat{C}_t)_{ij} N_{ij}(t)}{\sum_{(i,j) \in I} \mathbb{1}\{(\hat{C}_t^=)_{ij} = 1\} N_{ij}(t)}.$$

ENTRIES BELOW $1/2$

$$q_t = \frac{\sum_{(i,j) \in I} \mathbb{1}\{(\hat{C}_t^{\neq})_{ij} = 1\} (\hat{C}_t)_{ij} N_{ij}(t)}{\sum_{(i,j) \in I} \mathbb{1}\{(\hat{C}_t^{\neq})_{ij} = 1\} N_{ij}(t)}.$$

EMPIRICAL MATRIX $\hat{C}_t$

	1	2	3	4	5	6
1	1	.78	.34	.22	.47	.18
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AFTER THRESHOLDING: $\hat{C}_t^=$

	1	2	3	4	5	6
1	1	1	0	0	0	0
2	1	1	0	0	0	1
3	0	0	1	1	1	0
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Solution: Project $\hat{C}_t$ onto $\mathcal{C}$ using $\text{Proj}_{\mathcal{C}}$, and set $C_t = \text{Proj}_{\mathcal{C}}(\hat{C}_t)$.

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STEP 4: FORM THE PROJECTED INSTANCE C_t

$$C_t = p_t \tilde{C}_t + q_t (\mathbf{1} - \tilde{C}_t).$$

AFTER THRESHOLDING: $\hat{C}_t^=$

	1	2	3	4	5	6
1	1	1	0	0	0	0
2	1	1	0	0	0	1
3	0	0	1	1	1	0
4	0	0	1	1	1	0
5	0	0	1	1	1	0
6	0	1	0	0	0	1

$$\hat{C}_t^{\neq} = \mathbf{1} - \hat{C}_t^= \text{entrywise}$$

- **Unique optimizer:**

$$\lambda^*(\sigma; C_t) = \arg \max_{\lambda \in \mathcal{P}_I} \left\{ \inf_{C' \in \text{MIN}(C_t)} \sum_{(i,j) \in I} \lambda_{ij} \text{KL}((v_t)_{ij}, v'_{ij}) - \frac{\sigma}{2} \|\lambda\|^2 \right\}.$$

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- **Positive exploration:**

$$\lambda_\epsilon^*(\sigma; C_t) = (1 - \epsilon) \lambda^*(\sigma; C_t) + \epsilon \text{Unif}(I).$$

- **Unique optimizer:**

$$\lambda^*(\sigma; C_t) = \arg \max_{\lambda \in \mathcal{P}_I} \left\{ \inf_{C' \in \text{MIN}(C_t)} \sum_{(i,j) \in I} \lambda_{ij} \text{KL}((v_t)_{ij}, v'_{ij}) - \frac{\sigma}{2} \|\lambda\|^2 \right\}.$$

- **Positive exploration:**

$$\lambda_\epsilon^*(\sigma; C_t) = (1 - \epsilon) \lambda^*(\sigma; C_t) + \epsilon \text{Unif}(I).$$

- **D-tracking** (Garivier & Kaufmann, 2016)¹:

$$(i_{t+1}, j_{t+1}) \in \arg \max_{(i,j) \in I} \left\{ t (\lambda_\epsilon^*(\sigma; C_t))_{ij} - N_{ij}(t) \right\}.$$

¹ Garivier, A., & Kaufmann, E. (2016). Optimal best arm identification with fixed confidence. In V. Feldman, A. Rakhlin, & O. Shamir (Eds.), *Proceedings of the 29th Annual Conference on Learning Theory* (Vol. 49, pp. 998–1027). PMLR.

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$$(i_{t+1}, j_{t+1}) \in \arg \max_{(i,j) \in I} \left\{ t (\lambda_\epsilon^*(\sigma; C_t))_{ij} - N_{ij}(t) \right\}.$$

- Consequently,

$$\frac{N(t)}{t} \xrightarrow{\text{a.s.}} \lambda_\epsilon^*(\sigma; C).$$

¹ Garivier, A., & Kaufmann, E. (2016). Optimal best arm identification with fixed confidence. In V. Feldman, A. Rakhlin, & O. Shamir (Eds.), *Proceedings of the 29th Annual Conference on Learning Theory* (Vol. 49, pp. 998–1027). PMLR.

- A natural stopping rule is to use the GLR statistic

$$Z(t) = \inf_{C' \in \text{ALT}(C_t)} \sum_{(i,j) \in I} N_{ij}(t) \text{KL}((\hat{v}_t)_{ij}, v'_{ij}),$$

and to stop once $Z(t)$ has crossed a threshold $\beta(t, \delta)$:

$$\tau_\delta = \inf\{t \in \mathbb{N} : Z(t) > \beta(t, \delta)\}.$$

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Using the threshold in Kaufmann and Koolen (2021)¹,

Theorem (δ -correct Stopping Rule)

The GLR-based stopping rule is δ -correct, i.e.,

$$\Pr(\tau_\delta < \infty, [C_{\tau_\delta}] \neq [C]) \leq \delta.$$

¹ Kaufmann, E., & Koolen, W. M. (2021). Mixture martingales revisited with applications to sequential tests and confidence intervals. *Journal of Machine Learning Research*, 22(246), 1–44.

Theorem (Almost Asymptotically Optimal Algorithm)

Under D-tracking and the GLR-based stopping rule,

$$\limsup_{\delta \rightarrow 0} \frac{\mathbb{E}_C[\tau_\delta]}{\ln(1/\delta)} \leq \frac{1}{D_\epsilon^*(\sigma; C)},$$

where

$$D_\epsilon^*(\sigma; C) = \inf_{C' \in \text{ALT}(C)} \sum_{(i,j) \in I} (\lambda_\epsilon^*(\sigma; C))_{ij} \text{KL}(v_{ij}, v'_{ij}) - \frac{\sigma}{2} \|\lambda_\epsilon^*(\sigma; C)\|^2.$$

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- Information gathered under the optimal sampling rule

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Under D-tracking and the GLR-based stopping rule,

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- Regularization penalty

Theorem (Almost Asymptotically Optimal Algorithm)

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$$\lim_{\epsilon, \sigma \rightarrow 0} D_\epsilon^*(\sigma; C) = D^*(C).$$

- As $\epsilon, \sigma \rightarrow 0$, the modified hardness parameter converges to the optimal hardness $D^*(C)$

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$$\lim_{\epsilon, \sigma \rightarrow 0} D_\epsilon^*(\sigma; C) = D^*(C).$$

$$\frac{1}{D^*(C)} \leq \liminf_{\delta \downarrow 0} \inf_{\pi} \inf_{\delta\text{-correct}} \frac{\mathbb{E}_C[\tau_\delta(\pi)]}{\ln(1/\delta)} \leq \limsup_{\delta \rightarrow 0} \frac{\mathbb{E}_C[\tau_\delta]}{\ln(1/\delta)} \leq \frac{1}{D_\epsilon^*(\sigma; C)}.$$

ALTERNATIVE STOPPING RULE



భారతీయ పాఠశాల విజ్ఞాన పరిషత్ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

$$Z(t) = \inf_{C' \in \text{ALT}(C_t)} \sum_{(i,j) \in I} N_{ij}(t) \text{KL}((\hat{v}_t)_{ij}, v'_{ij})$$

- No closed-form expression for $Z(t)$

$$Z(t) = \inf_{C' \in \text{ALT}(C_t)} \sum_{(i,j) \in I} N_{ij}(t) \text{KL}((\hat{v}_t)_{ij}, v'_{ij})$$

- No closed-form expression for $Z(t)$
- **Workaround:** “Factoring out” $\min_{(i',j') \in I} N_{i'j'}(t)$:

$$\hat{Z}(t) = \left[\min_{(i',j') \in I} N_{i'j'}(t) \right] \left[\inf_{C' \in \text{ALT}(C_t)} \sum_{(i,j) \in I} \text{KL}((\hat{v}_t)_{ij}, v'_{ij}) \right].$$

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- The associated (alternative) stopping rule is

$$\hat{\tau}_\delta = \inf\{t \in \mathbb{N} : \hat{Z}(t) > \beta(t, \delta)\}.$$

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- The associated (alternative) stopping rule is

$$\hat{\tau}_\delta = \inf\{t \in \mathbb{N} : \hat{Z}(t) > \beta(t, \delta)\}.$$

Theorem (Alternative Stopping Rule is δ -Correct)

Because $Z(t) \geq \hat{Z}(t)$, the alternative stopping rule is still δ -correct.

Theorem (Near-Optimal Sample Complexity)

For any $\sigma > 0$, under D-tracking and the alternative stopping rule,

$$\limsup_{\delta \rightarrow 0} \frac{\mathbb{E}_C[\widehat{\tau}_\delta]}{\ln(1/\delta)} \leq \frac{1}{\tilde{D}_\epsilon^*(\sigma; C)},$$

where

$$\tilde{D}_\epsilon^*(\sigma; C) = \min_{(i,j) \in I} (\lambda_\epsilon^*(\sigma; C))_{ij} \inf_{C' \in \text{ALT}(C)} \sum_{(i,j) \in I} \text{KL}(v_{ij}, v'_{ij}) - \frac{\sigma}{2} \|\lambda_\epsilon^*(\sigma; C)\|^2.$$

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- Minimum sampling proportion

Theorem (Near-Optimal Sample Complexity)

For any $\sigma > 0$, under D-tracking and the alternative stopping rule,

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where

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- Overall KL separation from the alternatives

Theorem (Near-Optimal Sample Complexity)

For any $\sigma > 0$, under D-tracking and the alternative stopping rule,

$$\limsup_{\delta \rightarrow 0} \frac{\mathbb{E}_C[\widehat{\tau}_\delta]}{\ln(1/\delta)} \leq \frac{1}{\tilde{D}_\epsilon^*(\sigma; C)},$$

where

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- Small regularization penalty

Theorem (Near-Optimal Sample Complexity)

For any $\sigma > 0$, under D-tracking and the alternative stopping rule,

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A³CNP

Almost Asymptotically Optimal Active Clustering
with Noisy Pairwise Observations

- We define the **multiplicative suboptimality gap**:

$$\text{SG}_\epsilon(\sigma; C) = \frac{D_\epsilon^\star(\sigma; C)}{\tilde{D}_\epsilon^\star(\sigma; C)} = \frac{1/\tilde{D}_\epsilon^\star(\sigma; C)}{1/D_\epsilon^\star(\sigma; C)}$$

- We define the **multiplicative suboptimality gap**:

$$\text{SG}_\epsilon(\sigma; C) = \frac{D_\epsilon^*(\sigma; C)}{\tilde{D}_\epsilon^*(\sigma; C)} = \frac{\boxed{1/\tilde{D}_\epsilon^*(\sigma; C)}}{\boxed{1/D_\epsilon^*(\sigma; C)}}$$

Upper bound under the alternative stopping rule

Upper bound from the GLR-based stopping rule

- We define the **multiplicative suboptimality gap**:

$$\text{SG}_\epsilon(\sigma; C) = \frac{D_\epsilon^\star(\sigma; C)}{\tilde{D}_\epsilon^\star(\sigma; C)} = \frac{1/\tilde{D}_\epsilon^\star(\sigma; C)}{1/D_\epsilon^\star(\sigma; C)} \leq \underbrace{\frac{\overline{m}/\underline{m}}{1 - O_\epsilon(\sigma)}}_{\approx 1}$$

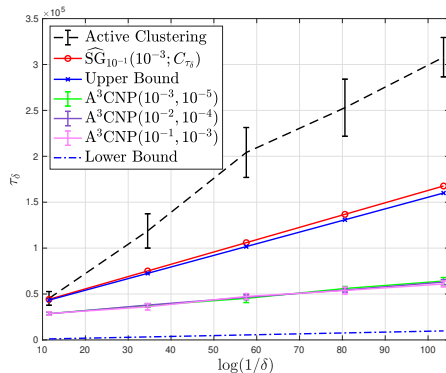
$$\overline{m} = \max_{(i,j) \in I} (\lambda_\epsilon^\star(\sigma; C))_{ij}, \quad \underline{m} = \min_{(i,j) \in I} (\lambda_\epsilon^\star(\sigma; C))_{ij}.$$

- We define the **multiplicative suboptimality gap**:

$$\text{SG}_\epsilon(\sigma; C) = \frac{D_\epsilon^\star(\sigma; C)}{\tilde{D}_\epsilon^\star(\sigma; C)} = \frac{1/\tilde{D}_\epsilon^\star(\sigma; C)}{1/D_\epsilon^\star(\sigma; C)} \leq \underbrace{\frac{\overline{m}/\underline{m}}{1 - O_\epsilon(\sigma)}}_{\approx 1}$$

- In practice, we use the **data-dependent proxy**:

$$\widehat{\text{SG}}_\epsilon(\sigma; C_t) = \frac{\max_{(i,j) \in I} (\lambda_\epsilon^\star(\sigma; C_t))_{ij}}{\min_{(i,j) \in I} (\lambda_\epsilon^\star(\sigma; C_t))_{ij}}.$$

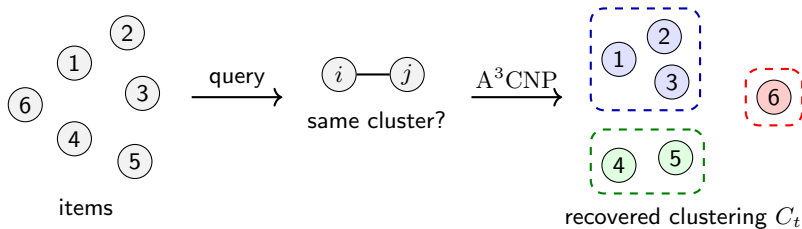


$$p = 0.6, q = 0.4$$

- A^3CNP has slope close to the lower bound slope
- $\widehat{SG}_\epsilon(\sigma; C_t)$ closely tracks the upper bound
- A^3CNP significantly improves over the Active Clustering algorithm of Chen et al. (2023)¹ as $\delta \rightarrow 0$

¹ Y. Chen, R. K. Vinayak, and B. Hassibi, "Crowdsourced clustering via active querying: Practical algorithm with theoretical guarantees," *Proceedings of the AAAI Conference on Human Computation and Crowdsourcing*, vol. 11, no. 1, pp. 27–37, Nov. 2023. [Online]. Available: <https://ojs.aaai.org/index.php/HCOMP/article/view/27545>

- We formulated noisy pairwise active clustering as a **fixed-confidence bandit problem**
- We derived an **instance-dependent lower bound** on the expected sample complexity
- We proposed A^3CNP (D-tracking + efficient stopping rule), and proved it is **near-optimal**



Full paper available at [arXiv:2602.05690](https://arxiv.org/abs/2602.05690)